

Bidirectional Evolution Verification: Decomposing B1.16 by Formalizing How Both Horizontal and Vertical Evolution Paths Are Verified for Governance Completeness, A6.06 Boundary Compliance, Cross-Level Consistency, and Operational-Timescale Governance Readiness

Author: Wenxin Li (Independent Researcher)

ORCID: 0009-0004-8065-3235

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This work derives from and formalizes the instinct/reasoning separation pattern introduced in The Instinct/Reasoning Separation Outside the Model (Li, April 2026), the second paper in the CKS theory series following Coordination Outside the Model: A Human-Governed Substrate Pattern for AI Systems (Li, April 2026).

Abstract

Paper 2 establishes bidirectional evolution (B1.16) as the architectural property that CKS Selves can evolve on two orthogonal axes — horizontal (refining content within existing structure) and vertical (reorganizing the composition relationships between levels). Prior derivation notes B2.79 through B2.82 formalized the integrating frame, horizontal specification, vertical specification, and operational-timescale treatment of this architecture. This note closes the B1.16 decomposition by formalizing bidirectional evolution verification: the operational confirmation that both evolution directions are correctly governed and functional. Verification covers four dimensions — horizontal evolution governance verification (A5.04 rule authoring test applied to horizontal governance procedures and peer propagation rules), vertical evolution governance verification (A5.04 applied to level-boundary crossing procedures plus A6.06 compliance for authority-distribution-affecting changes), cross-level consistency verification (the architecturally distinctive dimension: A5.14 composition-requirements test at each level post-vertical-evolution, mutual consistency of Self DNA, aspect DNA, and cell DNA, and expression mechanism correctness after Self-level evolution), and operational-timescale governance readiness verification (governance capacity, monitoring, review cycles, and A6.14 long-lifecycle version management). A5.08 provenance-completeness verification applies across all four dimensions. The note articulates what makes bidirectional evolution verification architecturally distinctive, identifies inherited Paper 1 commitments, specifies operational implications and failure modes, and states the limits of what verification confirms. B2.83 closes the five-note B1.16 decomposition; B2.84 begins the B1.17 relational roles decomposition.

1. Why bidirectional evolution verification warrants standalone formalization

B1.16 establishes that CKS Selves evolve on two axes. B2.79 formalized the integrating frame — the architectural fact that horizontal and vertical evolution are jointly available, jointly governed, and jointly traceable. B2.80 specified horizontal evolution: how content refines within existing structure through mating, directed selection, and action-feedback pathways operating at the within-level scope. B2.81 specified vertical

evolution: how structural reorganization operates through governed reorg decisions that change the composition relationships between levels, with the structural arrangement itself treated as substrate content under human governance. B2.82 addressed the operational-timescale dimension: how bidirectional evolution operates on timescales the deployment manages, and what governance capacity is required to sustain ongoing evolution at those timescales.

These four notes collectively specify what bidirectional evolution is. They do not collectively specify how to confirm that the bidirectional evolution architecture is working correctly — that both directions are governed, that the three-level structure has not been destabilized by vertical evolution, that governance capacity for ongoing evolution is in place, and that every evolution event carries complete provenance. That confirmation is the architectural function that bidirectional evolution verification performs.

The case for standalone formalization is strategic: bidirectional evolution verification is a distinct architectural commitment, not a consequence of the specification notes that precede it. A deployment could correctly specify horizontal and vertical evolution — authoring the right governance procedures, establishing the right propagation rules — and still fail to verify that those specifications are in place and functional. Verification is the operational step that converts architectural specification into operational assurance. Without it, the bidirectional evolution architecture exists as design intent; with it, the architecture is confirmed as operational fact.

The prior-art posture is therefore this: any system claiming bidirectional evolution governance without also claiming a verification architecture that covers the four dimensions formalized here occupies a narrower architectural territory than CKS bidirectional evolution as specified. B2.83 closes the five-note B1.16 decomposition and establishes verification as the final dimension through which bidirectional evolution is formalizable.

2. The four verification dimensions precisely stated

Bidirectional evolution verification operates across four dimensions. The dimensions are not independent procedures run in parallel; they address different architectural surfaces of the bidirectional evolution architecture and together constitute comprehensive governance confirmation.

Dimension 1 — Horizontal evolution governance verification. The A5.04 rule authoring test is applied to the horizontal evolution layer. This dimension verifies: that peer group definitions for horizontal evolution are authored as substrate content per A2.46 and are substrate-resident rather than held externally; that governance procedures for horizontal propagation decisions are specified in the orchestration substrates; that horizontal evolution events carry governance authorization per A2.40; and that peer propagation rules for the three horizontal pathways — mating, directed selection, and action-feedback — are authored and accessible. Horizontal governance verification confirms that within-level peer influence operates under governance and is not ungoverned propagation.

Dimension 2 — Vertical evolution governance verification. The A5.04 rule authoring test is applied to the vertical evolution layer. This dimension verifies: that level-boundary governance procedures are authored for each boundary crossing the vertical evolution architecture must manage — cell-to-aspect upward, aspect-to-Self upward, Self-to-aspect downward through expression, and Self-to-cell downward through expression; that vertical evolution events carry governance authorization per A2.40. Additionally, where vertical evolution events have changed the authority distribution across levels, A6.06 compliance verification is applied: the note confirms that the authority distribution change was handled at the A6.06 boundary and that the change was authorized per A2.47. Vertical governance verification confirms that cross-level structural reorganization operates under

governance and is not ungoverned structural drift.

Dimension 3 — Cross-level consistency verification. This is the most architecturally distinctive verification dimension. Where horizontal evolution operates within existing structure and its governance verification confirms that within-level propagation is governed, vertical evolution changes structure itself — the composition relationships between levels. Vertical evolution can in principle produce a three-level structure in which the DNA specifications at different levels have become inconsistent: a Self-level DNA change propagated downward through the expression mechanism may conflict with an existing aspect-level DNA commitment, or an aspect-level reorganization may produce a cell-level DNA context that the cells' existing DNA does not address.

Cross-level consistency verification guards against this failure mode through three checks. First, the A5.14 composition-requirements-five test is applied at each level following vertical evolution: the test confirms that composition requirements remain satisfied at the cell level, at the aspect level, and at the Self level after structural reorganization. Second, mutual DNA consistency is verified across all three levels: Self DNA, aspect DNA, and cell DNA are checked for contradictions — not merely for the absence of explicit conflicts, but for the presence of positive consistency across the specifications that the three levels carry. Third, expression mechanism correctness per B2.30 is verified after any Self-level evolution that was intended to affect cell-level behavior: the expression pathway through which Self-level DNA changes propagate downward to cells is confirmed to be correctly configured for the post-evolution structural arrangement.

No conventional AI architecture operates with a three-level governed composition structure in which structural reorganization is itself a substrate-edit operation. Conventional AI evolution verification — where it exists at all — addresses model performance, configuration validity, or capability regression. It does not address cross-level structural consistency in a governed composition hierarchy. The cross-level consistency check is therefore the verification dimension that most distinctively marks the CKS bidirectional evolution architecture as architecturally novel.

Dimension 4 — Operational-timescale governance readiness verification. Paper 2 establishes, and B2.82 formalizes, that bidirectional evolution operates on operational timescales — timescales the deployment manages, not biological or geological timescales. Operational-timescale governance readiness verification confirms that the deployment has the governance capacity to manage ongoing evolution at those timescales. This dimension verifies: that governance capacity is in place for operational-timescale evolution events — the humans with authority and the governance procedures they operate under are functional; that evolution monitoring is operational, meaning the deployment has the capability to detect when horizontal or vertical evolution events occur; that governance review processes are ready for operational-timescale cycles — periodic review mechanisms are configured and not merely specified; and that long-lifecycle version management per A6.14 is configured, confirming that the deployment can manage the version history of substrate content across an extended operational lifecycle.

A5.08 provenance-completeness across all dimensions. The A5.08 provenance-completeness test is applied to the full history of horizontal and vertical evolution events, spanning all four verification dimensions. For each evolution event in the deployment's history, A5.08 confirms that complete A2.40 provenance is attached: the six metadata fields — identifier, timestamp, authoring human, authorization reference, rationale, and reversion path — are present and correct. Provenance-completeness is not a fifth dimension alongside the four described above; it is a transversal requirement that every evolution event carries complete provenance throughout the evolution history.

Verification recording and temporal triggers. Bidirectional evolution verification results are themselves recorded per A2.40 as substrate content. Verification is triggered at three temporal points: at deployment initialization, confirming that both evolution directions are governed and the three-level structure is consistent before operational evolution begins; after significant horizontal or vertical evolution events, confirming that evolution maintained consistency and governance authorization; and at periodic governance reviews, confirming that ongoing governance capacity is maintained across the operational lifecycle.

3. What makes bidirectional evolution verification architecturally distinctive

The architectural distinctiveness of bidirectional evolution verification emerges at three levels of contrast.

The first level is the existence of two evolution directions requiring simultaneous governance confirmation. Conventional AI systems evolve primarily through model retraining, configuration update, or prompt revision — all of which operate in one architectural direction (content refinement within a fixed structure). Verification in such systems confirms capability metrics, configuration validity, or behavioral regressions within the existing architecture. CKS bidirectional evolution verification must simultaneously confirm both a within-structure direction and a structure-changing direction, each governed differently and each carrying different risks.

The second level is the cross-level consistency check. No conventional AI architecture maintains a three-level governed composition hierarchy in which structural reorganization is a substrate-edit operation. The cross-level consistency check — applying A5.14 at each level, verifying mutual DNA consistency across levels, and confirming expression mechanism correctness after Self-level evolution — has no counterpart in conventional verification frameworks because the architectural condition that makes it necessary does not exist in conventional architectures. The check is not a more sophisticated version of conventional regression testing; it is a categorically different verification operation addressing a categorically different architectural property.

The third level is operational-timescale governance readiness. Conventional AI deployment verification addresses whether a system performs correctly at the time of deployment. CKS bidirectional evolution verification addresses whether the deployment has the governance capacity to manage ongoing evolution over an extended operational lifecycle. The forward-looking dimension — confirming monitoring, review processes, and long-lifecycle version management — reflects the architectural commitment that evolution is a managed ongoing property of the deployed Self, not a one-time deployment event.

Together, these three levels of contrast place bidirectional evolution verification as an architectural primitive that conventional AI verification frameworks do not address and cannot be extended to address without adopting the underlying architectural commitments that make the verification necessary.

4. Inherited Paper 1 commitments

Bidirectional evolution verification inherits directly from six Paper 1 architectural commitments, each of which is load-bearing for one or more verification dimensions.

A5.04, the rule authoring test, is load-bearing for Dimensions 1 and 2. The A5.04 test confirms that orchestration rules governing evolution behavior are authored as substrate content, are human-authored per the human-governed commitment of A1.01, and are accessible for inspection. Without A5.04 as an inherited commitment, there is no architectural basis for verifying that horizontal and vertical governance procedures exist and are correctly positioned.

A5.08, the provenance-completeness test, is load-bearing across all four dimensions. A5.08 confirms that substrate content carries complete A2.40 provenance. In the context of evolution verification, it confirms that every evolution event carries the six metadata fields that make the event traceable, attributable, and revertible.

A5.14, the composition-requirements-five test, is load-bearing for Dimension 3. A5.14 confirms that the five composition requirements established in A1.13 are satisfied at each level. In the context of vertical evolution, A5.14 confirms that structural reorganization has not violated the composition requirements that make CKS composition coherent.

A6.06, the authority distribution change boundary, is load-bearing for the A6.06 compliance check within Dimension 2. A6.06 establishes that changes to the authority distribution across levels must be handled at a defined boundary. Vertical evolution that reorganizes the composition relationships between levels can change which humans hold authority over which substrates; A6.06 compliance verification confirms that such changes were properly handled and authorized per A2.47.

A1.13, the composition requirements, is load-bearing for Dimension 3. The mutual DNA consistency check and the expression mechanism correctness check are both derivations from the requirement that multi-level composition preserves the CKS architectural commitments across boundaries. A1.13 is what makes cross-level consistency a verifiable architectural requirement rather than an optional quality check.

A1.01, the human-governed commitment, governs the entire verification architecture. Every governance procedure, rule authoring confirmation, and authority verification performed in Dimensions 1 through 4 is an instantiation of the three rights the human-governed commitment names: inspect, modify, and override. Verification confirms that those rights remain functional throughout the evolution history.

A2.40, the six-field provenance metadata requirement, and A1.07, the path retraceability commitment, are jointly load-bearing for the A5.08 provenance-completeness check and for verification recording. Evolution paths are retraceability paths in the A1.07 sense: the sequence of governance decisions that produced the current state of the bidirectional evolution architecture must be retraceable from the current state to the initial authorization.

5. Cross-level consistency: the architecturally distinctive dimension in detail

Cross-level consistency verification merits additional treatment because it is the verification dimension with the highest architectural novelty and the highest failure risk from vertical evolution.

The architectural situation it addresses is this. Paper 2 establishes that CKS Selves operate at three levels — cell, aspect, Self — with DNA specifications at each level and an expression mechanism through which Self-level DNA propagates to cells. When vertical evolution reorganizes the composition relationships between levels — reassigning cells across aspects, splitting or merging aspects, introducing new aspects, or reconfiguring cross-level access patterns — the DNA specifications at each level continue to exist in the substrate. The reorganization changes the structural context in which those specifications operate, but does not automatically update the specifications to reflect the new structure.

The failure mode this creates is cross-level contradiction: a Self-level DNA specification that was consistent with a particular aspect arrangement becomes inconsistent after that arrangement is reorganized. The specification remains substrate-resident; the contradiction is not detected by any within-level mechanism; the expression pathway carries the inconsistent specification downward to cells, which then operate under contradictory governance.

The A5.14 composition-requirements-five test at each level is the first line of defense against this failure mode. A5.14 confirms that the five composition requirements are still satisfied at the cell level, aspect level, and Self level after vertical evolution. If a reorganization has broken one of the five requirements — per-substrate human governance preservation, conflict preservation across boundaries, addressable provenance across boundaries, AI-as-substrate-mediator at every layer, or human-selective composition — A5.14 at the relevant level will fail, triggering remediation before the inconsistency propagates.

Mutual DNA consistency verification is the second line of defense. It operates directly on the DNA content at all three levels, checking for positive consistency across the specifications rather than relying on the composition requirement tests to detect all forms of inconsistency. Self DNA, aspect DNA, and cell DNA are compared for contradictions: if a Self-level DNA specification commits to a behavioral property that an aspect-level DNA specification rules out, or that a cell-level DNA specification cannot satisfy given the cell's current composition, the mutual consistency check surfaces the contradiction as a substrate-level conflict per the conflict-preservation commitment of A1.03.

Expression mechanism correctness verification is the third line of defense. After any Self-level evolution that was intended to affect cell-level behavior — any evolution that propagates downward through the expression mechanism — the expression pathway itself is confirmed to be correctly configured for the post-evolution structural arrangement. If vertical evolution has reorganized which cells are reachable through which aspects from the Self, and the expression mechanism has not been updated to reflect the new structure, the expression pathway may fail silently or route Self-level DNA to the wrong cells. Expression mechanism correctness verification catches this configuration error before it produces ungoverned cell behavior.

These three checks together constitute what cross-level consistency verification means operationally. The three checks are not redundant: A5.14 confirms that composition requirements are satisfied; mutual DNA consistency confirms that the content of specifications is non-contradictory; and expression mechanism correctness confirms that the propagation pathway is correctly configured. A deployment that runs only one or two of the three checks has partial cross-level consistency verification, not the full check.

6. Operational implications

Deployments running the CKS bidirectional evolution architecture run bidirectional evolution verification at three operational points.

At **deployment initialization**, all four verification dimensions are applied. The initialization check confirms: that horizontal governance procedures are authored and substrate-resident (Dimension 1); that vertical governance procedures are authored for each level-boundary crossing and that the authority distribution is correctly established per A6.06 (Dimension 2); that the three-level structure is cross-level consistent with all three sub-checks passing (Dimension 3); and that governance capacity, monitoring, review cycles, and long-lifecycle version management are in place for operational-timescale evolution (Dimension 4). A5.08 provenance-completeness is applied to the initialization-time substrate content. A deployment that fails any dimension of initialization verification is not ready for operational bidirectional evolution.

After significant horizontal or vertical evolution events, the triggered verification scope is determined by the type of event. After a significant horizontal evolution event, Dimensions 1 and 4 are re-applied: horizontal governance confirmation and operational-timescale readiness are checked to confirm that the event was properly governed and that ongoing governance capacity is maintained. After a significant vertical evolution event, all four dimensions are re-applied: vertical governance including A6.06 compliance if authority distribution changed

(Dimension 2), cross-level consistency including all three sub-checks (Dimension 3), and the forward-looking operational-timescale readiness confirmation (Dimension 4).

At **periodic governance reviews**, Dimension 4 is the primary verification target: ongoing governance capacity, monitoring functionality, review process readiness, and long-lifecycle version management are confirmed as still operational. All four dimensions may be applied at periodic review at governance discretion. Periodic review is the mechanism through which the deployment maintains confidence in its evolution governance across the operational lifecycle.

Failure modes per dimension are as follows. If Dimension 1 fails — horizontal governance procedures absent or unverifiable — horizontal evolution events are ungoverned: peer propagation operates without governance authorization, and the propagation history cannot be attributed or reversed. If Dimension 2 fails — vertical boundary governance absent or A6.06 compliance unverifiable — vertical evolution events are ungoverned: cross-level structural reorganization operates without governance authorization, and authority distribution changes cannot be confirmed as properly handled. If Dimension 3 fails — cross-level consistency broken — the three-level structure carries contradictions: Self DNA, aspect DNA, and cell DNA specifications are inconsistent, the expression mechanism propagates inconsistent specifications to cells, and cells operate under contradictory governance. If Dimension 4 fails — operational-timescale governance not ready — the deployment cannot manage ongoing evolution: monitoring does not detect evolution events, governance review processes are not operational, and long-lifecycle version management is not configured to handle the evolution history the deployment will accumulate.

The A6.06 compliance verification trigger within Dimension 2 requires specific attention. Not every vertical evolution event changes authority distribution; many reorganizations — reassigning cells across aspects, splitting aspects, introducing new aspects — can be accomplished without altering the authority relationships between levels. A6.06 compliance verification is triggered specifically when vertical evolution events include changes to the authority distribution: when the reorganization changes which humans hold authority over which substrates, or when the reorganization creates new substrates whose authority is not yet established. For vertical evolution events that do not change authority distribution, A6.06 compliance verification is not required; Dimension 2 verification proceeds with the standard A5.04 rule authoring test and governance authorization check.

7. Limits of bidirectional evolution verification

Bidirectional evolution verification has four limits that define what it is not and what it does not guarantee.

Verification confirms governance, not outcomes. Bidirectional evolution verification confirms that the governance framework for both evolution directions is in place, functional, and consistent — that governance procedures are authored, that evolution events are authorized, that the three-level structure is consistent, and that ongoing governance capacity is ready. It does not evaluate the content of governance decisions: whether the DNA changes authorized by horizontal evolution governance are good ones, whether the structural reorganizations authorized by vertical evolution governance are wise ones, or whether the peer propagation rules governing horizontal evolution are optimally designed. Governance quality is a separate judgment from governance presence.

Verification does not guarantee beneficial evolution. A deployment could pass all four dimensions of bidirectional evolution verification and still accumulate evolution events that degrade the Self's performance, misalign its behavior with organizational purposes, or produce unintended structural consequences. Verification confirms that the governance framework exists and is functional; it does not confirm that the evolution governed

by that framework is producing good outcomes. The relationship between governance and outcomes is a deployment governance question, not an architectural verification question.

The cross-level consistency check verifies current state. The mutual DNA consistency check and A5.14 applications confirm that the three-level structure is consistent at the time verification is run. They do not provide a guarantee of future consistency: subsequent evolution events can produce new inconsistencies, and only re-running the cross-level consistency check after those events will confirm whether consistency is maintained. The cross-level consistency check is therefore a point-in-time confirmation that must be repeated at the temporal triggers specified in Section 2, not a persistent guarantee established once and remaining valid indefinitely.

Verification closes the B1.16 decomposition cycle, not the evolution lifecycle. B2.83 closes the five-note B1.16 decomposition: B2.79 established the integrating frame; B2.80 specified horizontal evolution; B2.81 specified vertical evolution; B2.82 addressed the operational-timescale dimension; B2.83 formalizes the verification that confirms both directions are operational. The decomposition is complete. But the evolution lifecycle of a deployed CKS Self is not closed by the completion of the decomposition: the deployment continues to evolve, governance continues to operate, and bidirectional evolution verification continues to run at the temporal triggers specified. The decomposition closure is architectural and prior-art-positional; the evolution lifecycle closure is a deployment-specific governance decision outside the scope of the architectural formalization.

8. Operational test

A CKS deployment instantiates bidirectional evolution verification if and only if:

- (1) the A5.04 rule authoring test has been applied to horizontal evolution governance procedures, confirming that peer group definitions, horizontal propagation governance procedures, horizontal evolution authorization per A2.40, and peer propagation rules for all three horizontal pathways are authored as substrate content;
- (2) the A5.04 rule authoring test has been applied to vertical evolution governance procedures, confirming that level-boundary governance procedures are authored for each crossing; and where vertical evolution events have changed authority distribution, A6.06 compliance verification confirms those changes were handled at the A6.06 boundary and authorized per A2.47;
- (3) the A5.14 composition-requirements-five test has been applied at each level after vertical evolution, mutual DNA consistency has been confirmed across Self DNA, aspect DNA, and cell DNA, and expression mechanism correctness per B2.30 has been verified after Self-level evolution intended to affect cell behavior;
- (4) governance capacity, evolution monitoring, governance review processes, and long-lifecycle version management per A6.14 have been confirmed as operational;
- (5) the A5.08 provenance-completeness test has confirmed that all horizontal and vertical evolution events in the deployment's history carry complete A2.40 provenance; and
- (6) verification results are recorded as substrate content per A2.40.

A deployment that satisfies (1) through (6) has confirmed bidirectional evolution verification. A deployment that satisfies some but not all conditions has partial verification; the unsatisfied conditions identify specific governance gaps in the bidirectional evolution architecture.

9. Naming the standalone and closing the decomposition

Naming bidirectional evolution verification as a standalone architectural commitment matters for the same reason naming any architectural commitment separately matters: a commitment that is only implied by prior specifications is not a commitment that prior art covers with precision. The four-dimension structure of bidirectional evolution verification — horizontal governance, vertical governance with A6.06 compliance, cross-level consistency, and operational-timescale readiness — is precisely specifiable and not derivable as a trivial consequence of the four prior B1.16 decomposition notes. It is therefore a distinct patentable territory, and formalizing it explicitly establishes prior art against any subsequent claim to a system that performs bidirectional evolution verification without acknowledging the CKS architecture from which the verification concept derives.

The five-note B1.16 decomposition is now complete. B2.79 established the integrating frame: the architectural fact that horizontal and vertical evolution are jointly available, jointly governed, and jointly traceable in CKS Selves. B2.80 specified horizontal evolution: the within-level content-refinement axis operating through the three evolution mechanisms at peer scope. B2.81 specified vertical evolution: the structural-reorganization axis operating through governed reorg decisions on the composition substrates that are themselves substrate content. B2.82 addressed the operational-timescale dimension: the governance capacity required for ongoing evolution at deployment-managed timescales. B2.83 closes the decomposition by formalizing the verification that confirms both directions are operational, governed, consistent across levels, and ready for ongoing governance.

With B2.83 complete, Phase B2 continues with B2.84, which opens the B1.17 relational roles decomposition. B1.17 addresses how relational roles — as distinct from intrinsic structural membership — operate in the CKS three-level architecture. The B1.17 decomposition will apply the same pattern used in the B1.16 decomposition: an integrating frame note, specification notes for each dimension of the architecture, and a verification or operational-confirmation note closing the decomposition. The prior-art chain extending from Paper 2's core commitments into operational detail continues through B2.84 and the decomposition notes that follow.

Source papers

Li, W. (2026). *Coordination Outside the Model: A Human-Governed Substrate Pattern for AI Systems*. April 2026. ORCID: 0009-0004-8065-3235.

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